

Rongzhe (Rob) Zhao

Health Data Science PhD Student | Biostatistics | Applied Machine Learning

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PROFILE

Health Data Science PhD student and published researcher working at the intersection of statistical modeling, machine learning, and applied AI. Experience spans clinical and public-health studies, predictive modeling, multimodal/Video-LLM evaluation and interpretability, and reproducible computing across AWS, Linux/HPC, and GPU/CUDA environments.

EDUCATION

George Washington University, Milken Institute School of Public Health | Washington, DC

PhD in Health Data Science, Biostatistics Track | Expected 2030

University of California San Diego | San Diego, CA

MS in Biostatistics | GPA: 3.67 | Jun 2025

University of California San Diego | San Diego, CA

BS in Statistics and Probability | GPA: 3.50 | Jun 2022

TECHNICAL SKILLS

Programming & Data: Python (object-oriented programming, data structures & algorithms), R, SQL, JavaScript, Bash, pandas, NumPy, SciPy, statsmodels, Jupyter Notebook/Lab

AI/ML: PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, Hugging Face Transformers, predictive modeling, deep learning, random forests, gradient boosting, SVMs, PCA/PLS, computer vision, torchvision, large language models (LLMs), neural-network and LLM fine-tuning, multimodal and Video LLMs, RAG, embeddings and vector databases, prompt engineering, LLM evaluation, model interpretability

Cloud, Systems & HPC: AWS (EC2, S3, IAM, Lambda, SageMaker, SNS, SQS, CloudWatch, AWS CLI), Docker, Linux/SSH, SLURM, HPC, GPU/CUDA computing, parallel computing

Data & Software Engineering: Git/GitHub (branches, merges, pull requests), Spark/PySpark, PostgreSQL, MySQL, SQLite, NoSQL, REST APIs, FastAPI/Flask, CI/CD, unit testing, Conda/pip/venv/renv, VS Code, RStudio

Statistics & Study Design: Regression/GLMs, mixed-effects and longitudinal models, survival analysis, Poisson/negative binomial models, causal inference, clinical-trial analysis, experimental design/A/B testing, Monte Carlo simulation, missing-data methods, power/sample-size analysis, cross-validation, hyperparameter tuning, class-imbalance methods

R, Visualization & Research: tidyverse/dplyr, data.table, lme4, survival, glmnet, caret/tidymodels, matplotlib, ggplot2, Tableau, dashboards/Shiny, LaTeX/Overleaf, R Markdown/Quarto

Languages: Mandarin (native), English (professional working proficiency)

RESEARCH EXPERIENCE

Data Discovery Lab, George Washington University | Washington, DC

Research Assistant | Advisor: Dr. Fang Jin | Dec 2025 - Present

From Behavioral Diagnosis to Decision Basis: A Survey of Interpretability in Video LLMs

Ongoing Survey Manuscript

- Organize multimodal/Video-LLM evaluation and interpretability methods by explanatory strength across behavioral diagnosis, evidence-level explanation, and mechanism-level explanation, with faithfulness and causal validity as cross-cutting criteria.
- Built a 61-paper literature map spanning 2023-2026 and systematically coded 48 core Video-LLM studies to characterize explanatory claims, supporting evidence, and open validation gaps.
- Maintain a reproducible Git-based manuscript workflow in LaTeX/Overleaf, including Bash tooling for packaging source files and figures.

AI-Based Commercial Motor Vehicle Safety Study | Jun 2024 - May 2025

Graduate Research Assistant

- Analyzed AI-derived, deidentified observations for a two-year FMCSA-funded study, estimating risky driving prevalence and associations with time of day, location, and vehicle type.
- Supported implementation of a variable message system delivering targeted, real-time safety messages based on detected driving behaviors.

Tc-99m Tilmanocept Dosimetry Investigation | Sep 2021 - 2025

Biostatistics Researcher

- Supported kidney cancer clinical-study design, protocols, statistical analysis plans, data validation, and preprocessing; developed reusable functions for simulation, sample size estimation, and power analysis.

PROFESSIONAL EXPERIENCE

Mobalytics | Remote

Summer Data Analyst | Jun 2021 - Aug 2021

- Modeled player-history data using linear regression and decision trees to identify demonstrated skills and traits; translated results into product recommendations and stakeholder-ready visualizations.
- Researched neural-network applications in esports analytics and proposed versatility and consistency metrics to enhance Gamer Performance Index algorithms.

Accttrue Technology Co., Ltd. | Beijing, China / Remote

Search Engine Optimization Intern | Oct 2020 - May 2021

- Evaluated traffic, monetization, conversion, keyword, and A/B-test performance; delivered forecasts, competitive analyses, and site-structure recommendations.

Accenture, Applied Intelligence Group | Remote

Summer Digital Marketing Analyst | Jul 2020 - Sep 2020

- Integrated multi-source campaign data, built reporting dashboards, and presented Google Analytics performance findings to management.

SELECTED PROJECTS

External Fixation to ORIF: Predictors of Time to Definitive Surgery | 2025-2026

Reproducible Statistical Analysis and Publication

- Built a reproducible R/renv pipeline for 126 orthopaedic-trauma patients using dplyr and Poisson, quasi-Poisson, and negative binomial models; formal overdispersion testing supported the negative binomial model (dispersion 4.39; likelihood-ratio $p < 0.001$; ΔAIC 177.5).
- Automated sensitivity analyses, post-hoc contrasts, and ggplot2 publication graphics; the analysis supported a 2026 *Foot & Ankle Orthopaedics* article.

Distributed SNP Annotation Web Application on AWS | 2026

Team Course Project

- Contributed to a team-developed Python/Flask web application deployed on Linux-based Amazon EC2, enabling users to upload SNP variant files, monitor annotation jobs, and download merged CSV results from Amazon S3.
- Supported deployment of an asynchronous annotation pipeline using Amazon SNS/SQS to fan out regional and effect-prediction tasks to EC2 workers, with DynamoDB status tracking and IAM role-based access to AWS resources.

Assessing the Diagnostic Impact of the Craft Story Test for MCI Prediction | Mar 2023

Graduate Research Project

- Built logistic-regression models on 46,475 observations and compared ROC/AUC performance with and without the Craft Story Test; quantified sensitivity and specificity decreases of 0.0324 and 0.015 when the test was excluded.

TaskFlow Full-Stack Task Application | 2026

Software Engineering Project

- Built and versioned with Git a Python/FastAPI REST API with Pydantic validation and SQLAlchemy/SQLite relational models; implemented CRUD endpoints for tasks, categories, and subtasks and connected them to a JavaScript/React interface.

PUBLICATIONS

O'Mara, A., Teehan, E., Fealy, A. W., Lee, S. M., **Zhao, R.**, & DeBritz, J. "Examining Delays in Definitive Fixation: An Association of Patient Demographics in Orthopaedic Trauma Requiring External Fixation." *Foot & Ankle Orthopaedics*. 2026;11(2):24730114261450079. doi:10.1177/24730114261450079.

Lin, T., **Zhao, R.**, Tu, S., Wu, H., Hui, Z., & Tu, X. "On Modelling Relative Risks for Longitudinal Binomial Responses: Implications from Two Dueling Paradigms." *General Psychiatry*. 2023;36:e100977. doi: 10.1136/gpsych-2022-100977.